Erik Rye

Assistant Professor, Johns Hopkins University

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Research Interests

Network security, privacy, and Internet measurement, with an emphasis on the security and privacy consequences of persistent network identifiers in IPv6, Wi-Fi, and Bluetooth.

Education

PhD, Computer Science, University of Maryland, College Park 2025
Advisor: Dave Levin
Dissertation: *Empirically Evaluating the Security and Privacy Implications of Persistent Network Identifiers*

MS, Computer Science, Naval Postgraduate School 2015

MS, Applied Mathematics, Naval Postgraduate School 2015

BS, Mathematics, US Naval Academy 2008

Academic Experience

Assistant Professor, Johns Hopkins University 2026–Present

Research Assistant Professor, Johns Hopkins University 2025–2026
Visiting Research Scientist, University of Maryland

PhD Candidate, University of Maryland 2022–2025

Master Instructor, US Naval Academy 2015–2018

Awards and Honors

Caspar Bowden Award for Outstanding Research in Privacy Enhancing Technologies 2025
Distinguished Reviewer, ACM Internet Measurement Conference (IMC) 2024, 2025
Best Paper Award, ACM Internet Measurement Conference (IMC) 2021
Computer Science Nominee, Clements Teaching Award, US Naval Academy 2018
Computer Science Nominee, Apgar Teaching Award, US Naval Academy 2017
Hopper Award for Excellence in Computer Science, Naval Postgraduate School 2015
Church Award for Excellence in Mathematics, Naval Postgraduate School 2015
Graduated with Distinction in Computer Science, Naval Postgraduate School 2015
Graduated with Distinction in Applied Mathematics, Naval Postgraduate School 2015
Graduated with Honors in Mathematics, US Naval Academy 2008

Peer-Reviewed Publications

[1] **Erik Rye**, Robert Beverly. Cleaning the NTP Pool: Detecting and Mitigating NTP-Sourced IPv6 Scanning. *ACM Conference on Computer and Communications Security (CCS)*, 2026. *To appear*.

[2] **Erik Rye**, Dave Levin, Robert Beverly. Where Have All the Firewalls Gone? Security Consequences of Residential IPv6 Transition. *ACM Conference on Computer and Communications Security (CCS)*, 2026. *To appear*.

[3] Chase Kanipe, **Erik Rye**, Dave Levin, Robert Beverly. 6SEVEN: System for EValuating IPv6 ENumeration Algorithms. *ACM Internet Measurement Conference (IMC)*, 2026.

[4] Spencer Ye, Chase Kanipe, Peter Ryan, **Erik Rye**. Reactive Constraint-Based Geolocation of Internet Hosts. *NewGeo*, 2026.

[5] Robert Beverly, **Erik Rye**. On Borrowed Time: Measurement-Informed Understanding of the NTP Pool's Robustness to Monopoly Attacks. *Network and Distributed System Security Symposium (NDSS)*, 2026. Acceptance rate: 16%.

[6] **Erik Rye**, Dave Levin. WikIPedia: Unearthing a 20-Year History of IPv6 Client Addressing. *Passive and Active Measurement Conference (PAM)*, 2026. Acceptance rate: 32%.

[7] Steven Su, **Erik Rye**, Dave Levin, Robert Beverly. Buy It Now, Track Me Later: Attacking User Privacy via Wi-Fi AP Online Auctions. *Privacy Enhancing Technologies Symposium (PETS)*, 2025.

Acceptance rate: 20%.

[8] Sadia Nourin, **Erik Rye**, Kevin Bock, Nguyen Phong Hoang, Dave Levin. Is Nobody There? Good! Globally Measuring Connection Tampering without Responsive Endhosts. *IEEE Symposium on Security and Privacy (Oakland)*, 2025. Acceptance rate: 15%.

[9] **Erik Rye**, Dave Levin. Surveilling the Masses with Wi-Fi-Based Positioning Systems. *IEEE Symposium on Security and Privacy (Oakland)*, 2024. Acceptance rate: 17%. **Caspar Bowden PET Award, 2025.**

[10] **Erik Rye**, Dave Levin. IPv6 Hitlists at Scale: Be Careful What You Wish For. *ACM SIGCOMM*, 2023. Acceptance rate: 20%.

[11] **Erik Rye**, Robert Beverly. IPvSeeYou: Exploiting Leaked Identifiers in IPv6 for Street-Level Geolocation. *IEEE Symposium on Security and Privacy (Oakland)*, 2023. Acceptance rate: 17%.

[12] Travis Mayberry, Ellis Fenske, Dane Brown, Jeremy Martin, Christine Fossaceca, **Erik Rye**, Sam Teplov, Lucas Foppe. Who Tracks the Trackers? Circumventing Apple's Anti-Tracking Alerts in the FindMy Network. *Workshop on Privacy in the Electronic Society (WPES)*, 2021. Acceptance rate: 34%.

[13] **Erik Rye**, Robert Beverly, kc claffy. Follow the Scent: Defeating IPv6 Prefix Rotation Privacy. *ACM Internet Measurement Conference (IMC)*, 2021. Acceptance rate: 28%. **Best Paper Award.**

[14] Ellis Fenske, Dane Brown, Jeremy Martin, Travis Mayberry, Peter Ryan, **Erik Rye**. Three Years Later: A Study of MAC Address Randomization in Mobile Devices and When It Succeeds. *Privacy Enhancing Technologies Symposium (PETS)*, 2021. Acceptance rate: 19%.

[15] **Erik Rye**, Jeremy Blackburn, Robert Beverly. Reading In-Between the Lines: An Analysis of Dissenter. *ACM Internet Measurement Conference (IMC)*, 2020. Acceptance rate: 25%.

[16] **Erik Rye**, Robert Beverly. Discovering the IPv6 Network Periphery. *Passive and Active Measurement Conference (PAM)*, 2020. Acceptance rate: 29%.

[17] Jeremy Martin, Douglas Alpuche, Kristina Bodeman, Lamont Brown, Ellis Fenske, Lucas Foppe, Travis Mayberry, **Erik Rye**, Brandon Sipes, Sam Teplov. Handoff All Your Privacy – A Review of Apple's Bluetooth Low Energy Continuity Protocol. *Privacy Enhancing Technologies Symposium (PETS)*, 2019. Acceptance rate: 22%.

[18] **Erik Rye**, Robert Beverly. Sundials in the Shade: An Internet-Wide Perspective on ICMP Timestamps. *Passive and Active Measurement Conference (PAM)*, 2019. Acceptance rate: 27%.

[19] Lucas Foppe, Jeremy Martin, Travis Mayberry, **Erik Rye**, Lamont Brown. Exploiting TLS Client Authentication for Widespread User Tracking. *Privacy Enhancing Technologies Symposium (PETS)*, 2018. Acceptance rate: 16%.

[20] Jeremy Martin, Travis Mayberry, Collin Donahue, Lucas Foppe, Lamont Brown, Chadwick Riggins, **Erik Rye**, Dane Brown. A Study of MAC Address Randomization in Mobile Devices and When It Fails. *Privacy Enhancing Technologies Symposium (PETS)*, 2017. Acceptance rate: 23%.

[21] Jeremy Martin, **Erik Rye**, Robert Beverly. Decomposition of MAC Address Structure for Granular Device Inference. *Annual Computer Security Applications Conference (ACSAC)*, 2016. Acceptance rate: 23%.

[22] **Erik Rye**, Justin P. Rohrer, Robert Beverly. Revisiting AS-Level Graph Reduction. *Workshop on Network Science for Communication Networks (NetSciCom)*, 2016. Acceptance rate: 24%.

[23] Benjamin Davis, Ralucca Gera, Gary Lazzaro, Bing Yong Lim, **Erik Rye**. The Marginal Benefit of Monitor Placement on Networks. *Conference on Complex Networks*, 2016.

[24] Jonathan Roginski, Ralucca Gera, **Erik Rye**. The Neighbor Matrix: Generalizing the Degree Distribution. *Journal of Combinatorial Mathematics and Combinatorial Computing (JCMCC)*, 2016.

Technical Reports

[1] **Erik Rye**. Visualizing MAC and IPv6 Address Allocations. 2023.

[2] **Erik Rye**, Robert Beverly. Customer Management DNS Resource Records. *IETF RFC 8567*, 2019.

[3] **Erik Rye**, Jeremy Martin, Robert Beverly. EUI-64 Considered Harmful. 2019.

[4] **Erik Rye**, Robert Beverly. SDN as Active Measurement Infrastructure. 2017.

[5] **Erik Rye**, Justin Rohrer. Graph Reduction for Emulated Network Experimentation. 2015.

Patents

[1] **Erik Rye**, Robert Beverly. Discovering and Clustering IPv6 Addresses by Hardware Identifiers. US Patent 12,021,829, 2024.

Teaching Experience

Network Security, Johns Hopkins University
Fall 2026

Independent Research, SY495/SY496, US Naval Academy
Spring 2017; Fall 2017

Computer Networks, IC322, US Naval Academy
Fall 2016 (2 sections, course coordinator); Fall 2017 (3 sections, course coordinator)

Computer Architecture, IC220, US Naval Academy
Spring 2016 (2 sections); Spring 2017 (2 sections); Spring 2018 (2 sections, course coordinator)

Systems Programming, IC221, US Naval Academy
Spring 2018

Introduction to Cybersecurity, SY110/SI110, US Naval Academy
Fall 2015 (3 sections); Fall 2016

Introduction to Web Design, SI200, US Naval Academy
Spring 2016 (course coordinator); Spring 2017 (course coordinator)

Talks

[1] **Erik Rye**. Surveilling the Masses with Wi-Fi Positioning Systems. *Black Hat USA*, 2024.

[2] **Erik Rye**. IPvSeeYou: Exploiting Leaked Identifiers in IPv6 for Street-Level Geolocation. *Institute for Defense Analyses Cybersecurity Developmental Test Cross-Service Working Group (CyberDT XSWG)*, 2022.

[3] **Erik Rye**, Robert Beverly. IPvSeeYou: Exploiting Leaked Identifiers in IPv6 for Street-Level Geolocation. *Black Hat USA*, 2021.

[4] **Erik Rye**, Robert Beverly. Toxicity in Two Online Platforms: Dissenter and GitHub. *Sonoma State University*, 2021.

[5] **Erik Rye**. SDN as Active Measurement Infrastructure. *CAIDA Active Internet Measurements Workshop (AIMS)*, 2017.

[6] **Erik Rye**. Emulated Router Inference Kit. *CAIDA Active Internet Measurements Workshop (AIMS)*, 2015.

[7] Jon Roginski, **Erik Rye**. Evaluating Structural Disruption in Adaptive Networks. *INFORMS Annual Meeting*, 2014.

Media Coverage

[1] Brian Krebs. Why Your Wi-Fi Router Doubles as an Apple AirTag. *Krebs on Security*, 2024.

[2] Paul Wagenseil. Your Wi-Fi Router Could Tell Everyone Where You Live—Here's What You Can Do About It. *Tom's Guide*, 2021.

[3] Max Eddy. Is Your Crummy Router Giving Away Your Location? *PC Mag*, 2021.

[4] Thomas Claburn. Latest Phones Are Great at Thwarting Wi-Fi Tracking. Other Devices, Not So Much – Study. *The Register*, 2021.

[5] Thomas Claburn. MAC Randomization: A Massive Failure that Leaves iPhones, Android Mobes Open to Tracking. *The Register*, 2017.

Advising

[1] Hannah Ohm and Yi Li. *An Analysis of Wi-Fi-Based Positioning Systems: Comparing Google to iOS*. MS in Security Informatics capstone project, Johns Hopkins University, 2025. Advisor.

[2] Kirstin Thordarson. *Analysis of EUI-64-Based Addressing and Associated Vulnerabilities*. MS thesis, Naval Postgraduate School, 2020. Second reader.

[3] Terrence Kvitchko. *Sunup: ICMP Timestamp Behaviors in Fingerprinting*. MS thesis, Naval Postgraduate School, 2019. Second reader.

Service

Program Committees

ACM Conference on Computer and Communications Security (CCS)	2026
Security and Privacy in Communication Networks (SecureComm)	2026
ACM Internet Measurement Conference (IMC)	2024, 2025
ACM Internet Measurement Conference (IMC), Shadow PC	2017

Journal Reviewing

IEEE Transactions on Mobile Computing	2025
IEEE/ACM Transactions on Networking	2020
IEEE Communications Letters	2020